14/05

Pure Virtual Function

A virtual function whose definition is assigned with zero, must redefined by the child class.

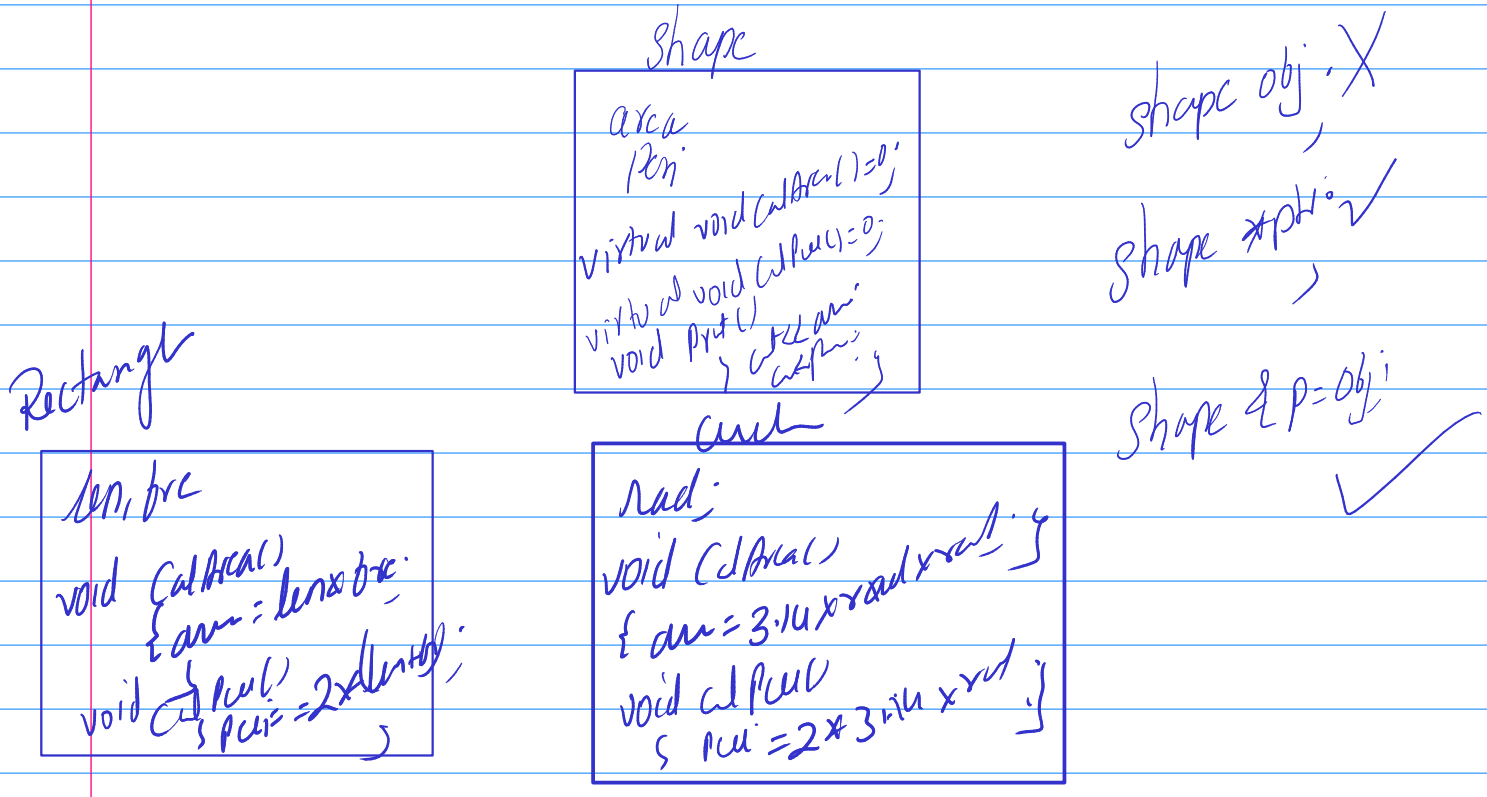
syntax :

virtual return type fun()=0;

If a class contains atleast one pure virtual function class becomes Abstract class .

For an abstract class we cannot create objects,but we can create pointers and references

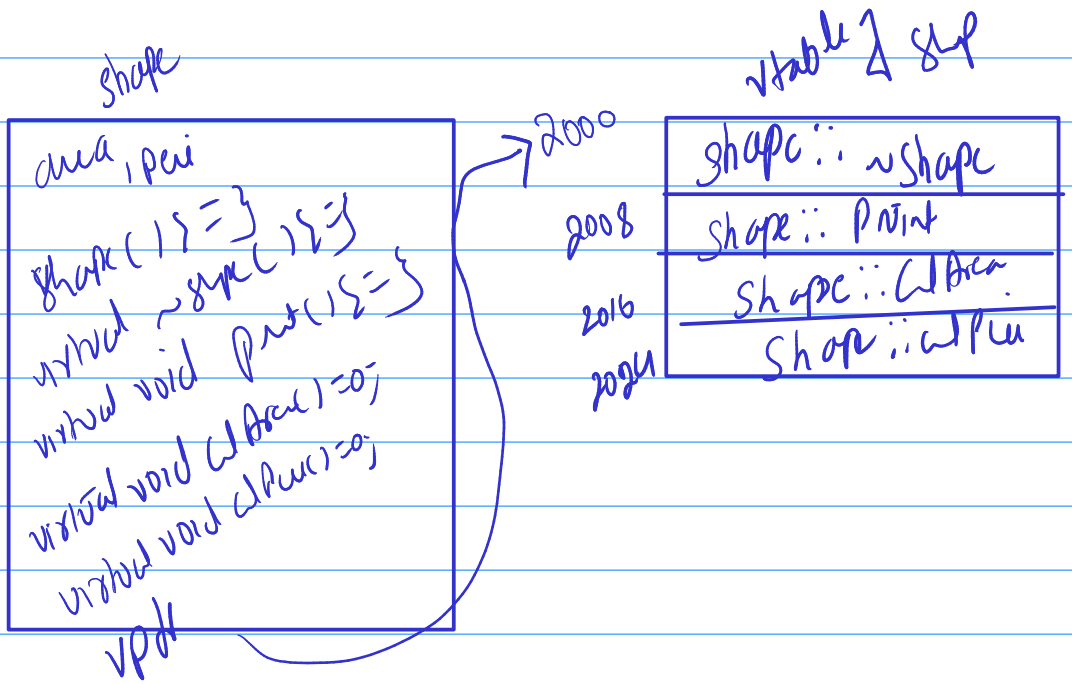
example of Abstract class



When a class contains virtual functions , compiler creates the virtual table at compile time and stores the virtual functions address in the virtual table.

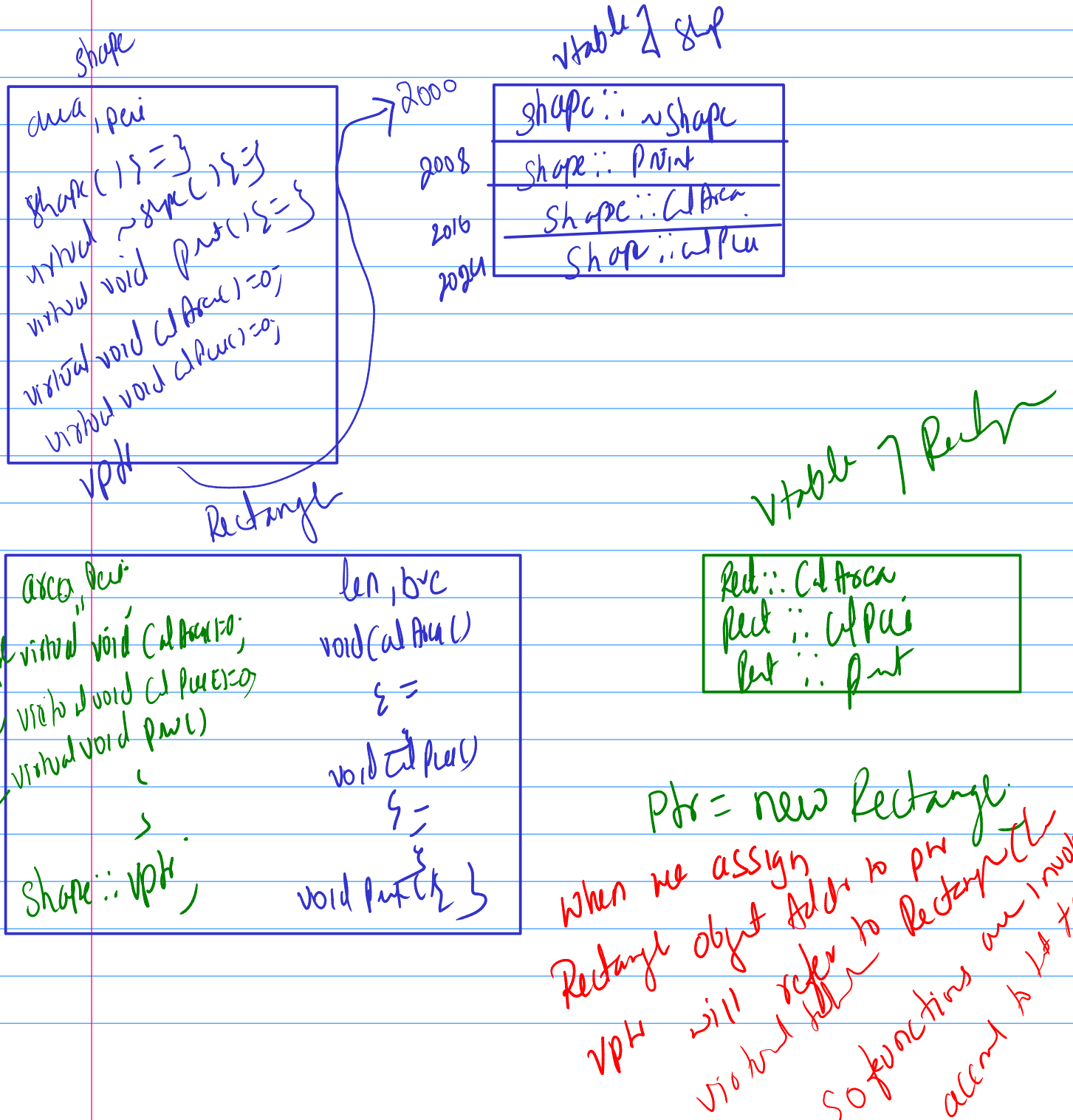
What is Virtual table?
A virtual table is a static array of function pointers. virtual table is also known as vtable.

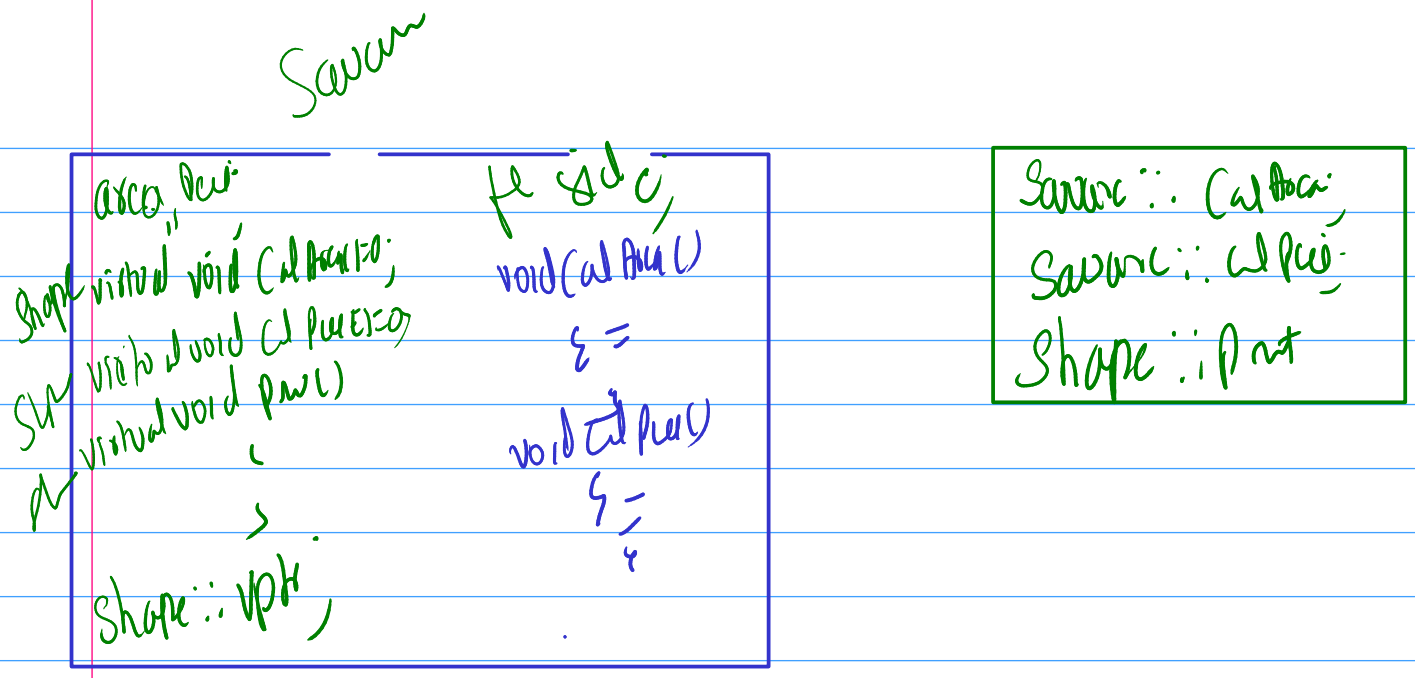
How connectivity created between the vtable and and class?
using virtual pointer, A virtual pointer is a hidden pointer supplied by the compiler as apublic data member to class, which holds the base address of the virtual table.



Only one virtual table will be created for a class(with virtul functions)
virtual table cannot be inherited.

vptr becomes a part of the object.
vptr can be inherited to child class.





is it virtual functions are faster or normal functions?
normal functions are faster